Example 12.7

Verify (i) closure property, (ii) commutative property, (iii) associative property, (iv) existence of identity, and (v) existence of inverse for following operation on the given set.

$m * n = m + n - mn;\ m, n \in \mathbb{Z}$

Solution

(i) The output $m + n - mn$ is clearly an integer and hence $*$ is a **binary operation** on $\mathbb{Z}$.

(ii) $m * n = m + n - mn = n + m - nm = n * m,\ \forall m, n \in \mathbb{Z}$. So $*$ has **commutative property.**

(iii) Consider $(m * n) * p = (m + n - mn) * p = (m + n - mn) + p - (m + n - mn)\, p$

$= m + n + p - mn - mp - np + mnp$ ... (1)

Similarly $m * (n * p) = m * (n + p - np) = m + (n + p - np) - m(n + p - np)$

$= m + n + p - np - mn - mp + mnp$ ... (2)

From (1) and (2), we see that $m * (n * p) = (m * n) * p$. Hence $*$ has **associative property.**

(iv) An integer e is to be found such that

$m * e = e * m = m,\ \forall m \in \mathbb{Z} \Rightarrow m + e - m\ e = m$

$\Rightarrow e\ (1 - m) = 0 \Rightarrow e = 0$ or $m = 1$. But m is an arbitrary integer and hence need not be equal to 1. So the only possibility is $e = 0$. Also $m * 0 = 0 * m = m,\ \forall m \in \mathbb{Z}$. Hence 0 is the identity element and hence the **existence of identity** is assured.

(v) An element $m' \in \mathbb{Z}$ is to be found such that $m * m' = m' * m = e = 0,\ \forall m \in \mathbb{Z}$.

$m * m' = 0 \Rightarrow m + m' - m\ m' = 0 \Rightarrow m' = \dfrac{m}{m-1}$. When $m=1$, m' is not defined.

When $m = 2$, m' is an integer. But except $m=2$, m' need not be an integer for all values of m. Hence **inverse does not exist in** $\mathbb{Z}$.

12.2.3 Some binary operations on Boolean Matrices

Definition 12.3

A **Boolean Matrix** is a real matrix whose entries are either 0 or 1.

Note that the boolean entries 0 and 1 can be defined in several ways. In electrical switch to describe "on and off", in graph theory, the "adjacency matrix" etc , the boolean entries 0 and 1 are used. We consider the same type of Boolean matrices in our discussion.

The following two kinds of operations on the collection of all boolean matrices are defined.

Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be any two boolean matrices of the **same type**. Then their **join** $\vee$ and **meet** $\wedge$ are defined as follows:

Definition 12.4: Join of A and B

$$A \vee B = [a_{ij}] \vee [b_{ij}] = [a_{ij} \vee b_{ij}] = [c_{ij}]$$

$$\text{where } c_{ij} = \begin{cases} 1, & \text{if either } a_{ij} = 1 \text{ or } b_{ij} = 1 \\ 0, & \text{if both } a_{ij} = 0 \text{ and } b_{ij} = 0 \end{cases}$$

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